

Shadman Salif Swanan

+880 1727-920359 | shadmansalif@gmail.com | Shadman Salif Swanan | Shad1573

Research Interests

Natural Language Processing, Large Language Models, Computer Vision, OCR, Deep Learning, Semantic Drift Analysis

Technical Skills

Programming Languages: Python, C++, JavaScript, PHP

Data Analysis: MySQL, Power BI, Microsoft Excel (Advanced), Process Modeling, EDA

AI/ML Frameworks: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Tensorflow, Hugging Face

Web & Tools: React, Vite, Node.js, Express.js, MongoDB, Git & GitHub, Tailwind CSS, Figma

Education

BRAC University <i>Bachelor of Science in Computer Science and Engineering</i>	2022 – 2026 CGPA: 3.84 / 4.00
Notre Dame College <i>Higher Secondary Certificate (Science)</i>	2018 – 2021 GPA: 5.00 / 5.00
Ideal School and College <i>Secondary School Certificate (Science)</i>	2008 – 2018 GPA: 5.00 / 5.00 (General Scholarship/ Dhaka Board)

Research Experience

Undergraduate Thesis Research BRAC University	2025 – 2026
Understanding Semantic Drift in Bengali: A Study Based on Diachronic Usage Patterns in Transformer Contexts – Engineered a comparative embedding framework using GloVe and transformer-based BanglaBERT with an Optuna-optimized hyperparameters across a 120-year corpus to detect true semantic drift with high data processing precision.	
Leaf Disease Research Independent	Apr 2026
– Developed TinyResNet, achieving 96.59% accuracy with distortion-aware augmentation techniques.	
Real-Time Fraud Detection System Independent	May 2026
– Built a fraud detection pipeline using LightGBM and developed a real-time transaction simulation console.	

Work Experience

Business Analyst Intern <i>Anwar Group of Industries</i>	Jun 2026 – Present Dhaka, Bangladesh
– Analyze sales and operational data to identify trends, generate business insights, develop analytical reports, and support data-driven decision-making and process improvement.	
GCI World Program <i>The University of Tokyo (Matsuo-Iwasawa Laboratory)</i>	Apr 2026 – July 2026 Remote
– Participated in an international AI education program focused on machine learning and data science.	

Projects

Shikkha.com React, Vite, Node.js, Express.js, MongoDB, JWT, Stripe, Tailwind CSS
– Developed a full-stack EdTech platform supporting course enrollment, online exams with automated grading, educator appointments, and e-commerce for books. Integrated JWT-based authentication, role-based access control, and Stripe Checkout.
Predictive Telecom Churn Optimization Framework Python, Scikit-learn, HistGradientBoosting, Pandas, EDA, Feature Engineering
– Engineered a machine learning pipeline processing 100K customer logs with an advanced Gradient Boosting model, achieving a 0.6900 ROC-AUC score and 64.00% churn recall to project an estimated 2.22M USD in annualized retention savings.
Token-Level POS Tagging Using RNN Models Python, TensorFlow, RNN, LSTM, GRU, BiLSTM, NLP
– Implemented and compared RNN, LSTM, GRU, and BiLSTM architectures for token-level part-of-speech tagging, evaluating sequence labeling performance across multiple recurrent neural network models.
Federal Flight Agency Mission-Critical Aerospace Network Design Cisco Packet Tracer, RIPv2, Static Routing, VLSM
– Designed a high-availability network for autonomous aircraft operations using VLSM and a hybrid routing strategy combining RIPv2 and static routes to ensure automatic failover and reliable communication.

Co-Curricular Activities

Vice President, Photography Dept, NDNSC | Champion, BRACU Intra-Spring 2022 Football Tournament.
Runners-Up in Crusade 2026, RS Football Tournament | Completed Stage Drama Course (BRACU RS).

References

Dr. Chowdhury Mofizur Rahman
Professor, Dept. of CSE
BRAC University, Dhaka, Bangladesh
Email: rahman.mofizur@bracu.ac.bd
Phone: +880 1715-152104

Dr. Farig Yousuf Sadeque
Associate Professor, Dept. of CSE
BRAC University, Dhaka, Bangladesh
Email: farig.sadeque@bracu.ac.bd
Phone: +880 1756-858357

Sk. Md. Masud Hasan
Manager-Taxation
Nestle Bangladesh PLC, Dhaka
Email: masud.hasan@bd.nestle.com
Phone: +880 1755-537114